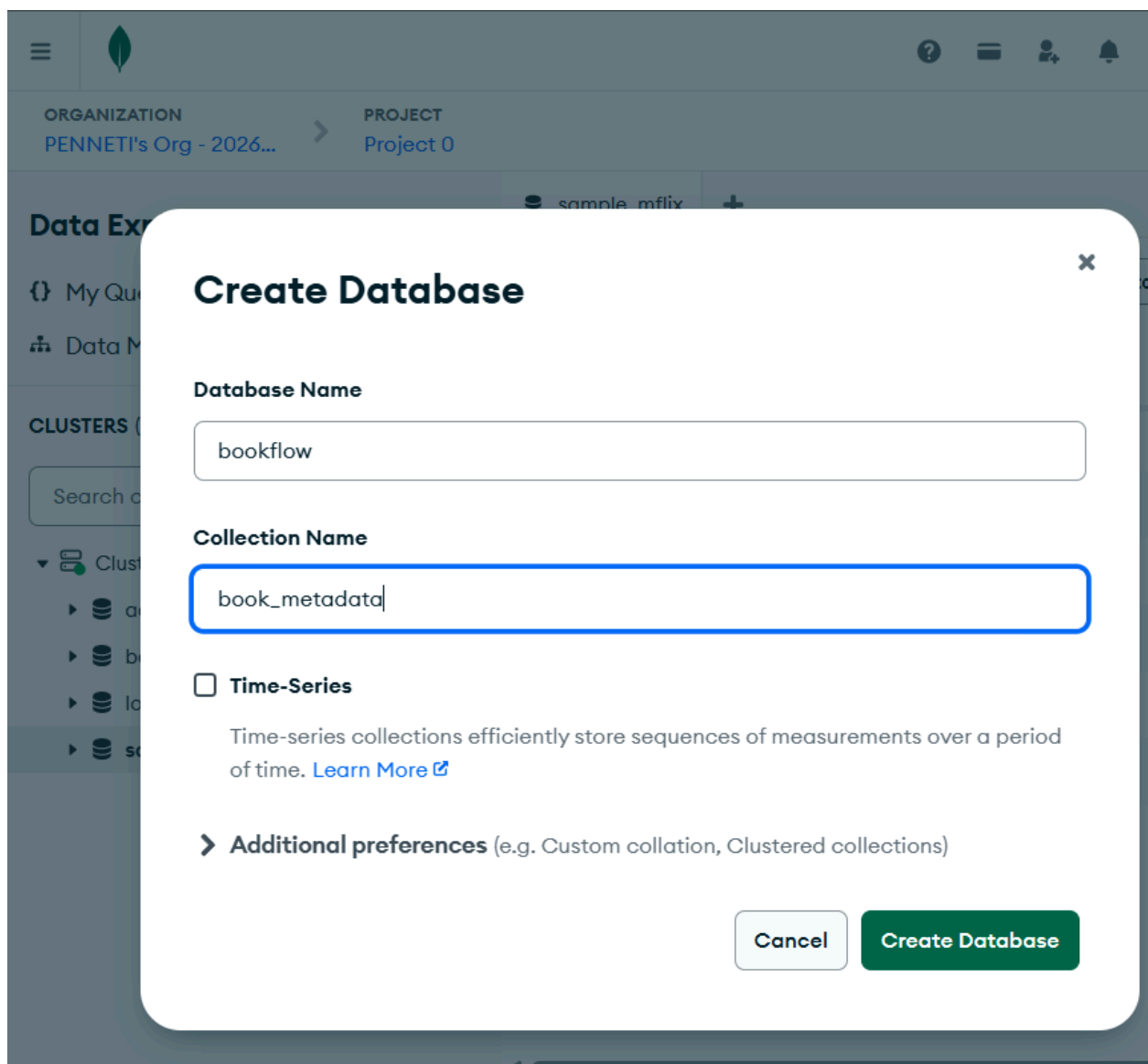


Week-3

Name: P.Bindhu Madhavi
Roll Number: 2520030077
Section: 7

STEP 1: Create Database



The screenshot shows a 'Create Database' modal dialog box overlaid on a dashboard. The dashboard background includes a sidebar with 'Data Explorer' and 'CLUSTERS' sections, and a top header with 'ORGANIZATION' (PENNETI's Org - 2026...) and 'PROJECT' (Project 0). The modal has a title bar with a close button (X). Inside, there are two text input fields: 'Database Name' with the value 'bookflow' and 'Collection Name' with the value 'book_metadata'. Below these is a checkbox for 'Time-Series' which is unchecked. A descriptive text for Time-Series says 'Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)'. At the bottom, there is a section for 'Additional preferences (e.g. Custom collation, Clustered collections)' and two buttons: 'Cancel' and 'Create Database'.

Create Database

Database Name

bookflow

Collection Name

book_metadata

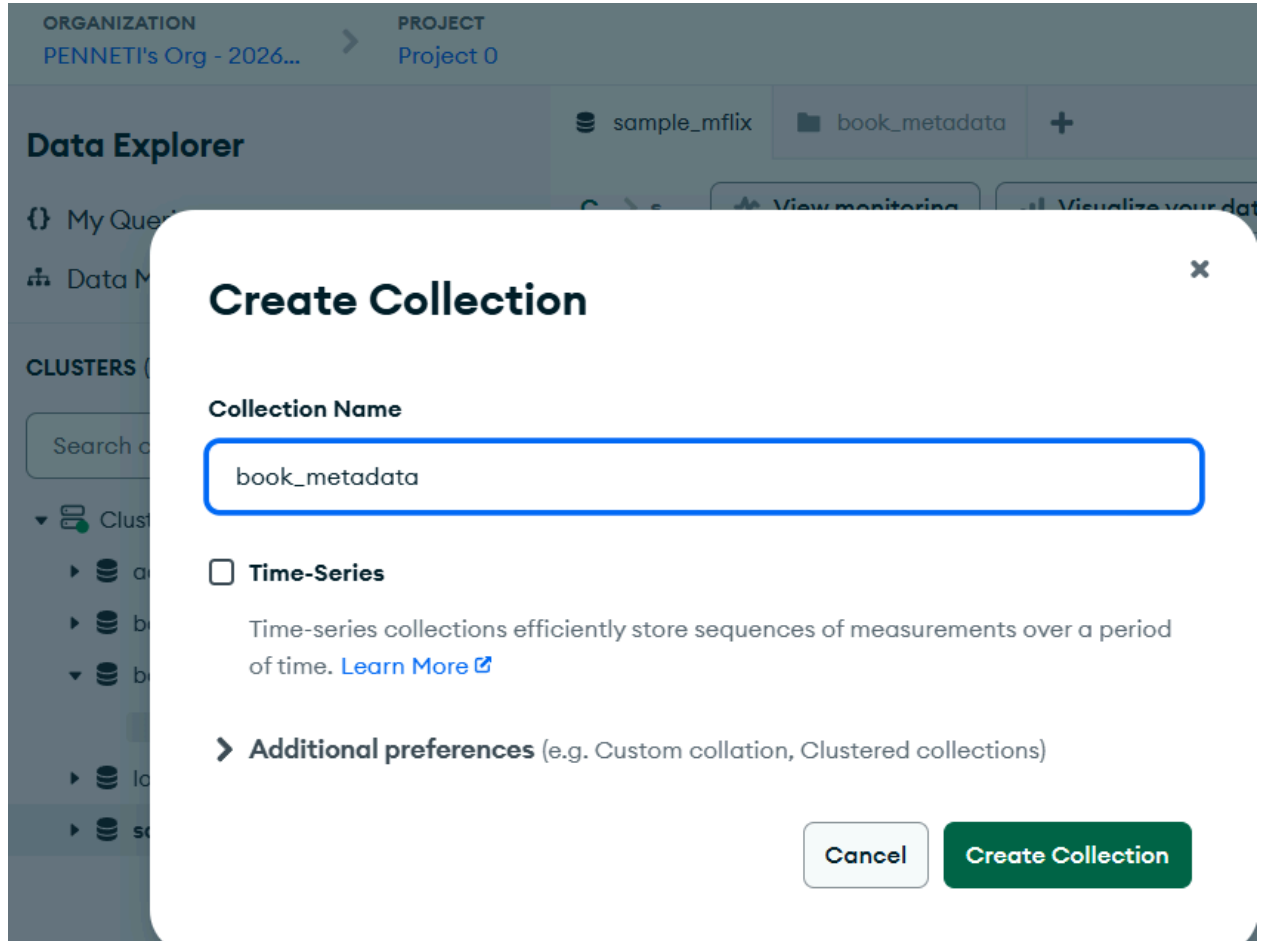
☐ **Time-Series**

Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

> Additional preferences (e.g. Custom collation, Clustered collections)

Cancel Create Database

STEP 2: Create Collection



The screenshot shows the MongoDB Data Explorer interface with a 'Create Collection' modal dialog open. The dialog has a title bar with a close button (X). Inside, the 'Collection Name' field is filled with 'book_metadata'. Below this, there is an unchecked checkbox for 'Time-Series' with a description: 'Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)'. Further down, there is a section for 'Additional preferences' with a right-pointing arrow and the text '(e.g. Custom collation, Clustered collections)'. At the bottom right of the dialog are two buttons: 'Cancel' and 'Create Collection'.

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

Data Explorer

sample_mflix book_metadata +

My Query

Data M

CLUSTERS (

Search a

Clust

Time-Series

Time-series collections efficiently store sequences of measurements over a period of time. [Learn More](#)

Additional preferences (e.g. Custom collation, Clustered collections)

Cancel Create Collection

STEP 3: Insert Document 1 in book_meta

Navigation bar: ORGANIZATION PENNETI's Org - 2026... PROJECT Project 0

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - bookflow
 - book_metadata**
 - local
 - sample_mflix

book_metadata +

Cluster0 > bookflow > book_metadata

View monitor

Documents 5 Aggregations Schema Indexes 1

Timeline: [Clock icon] [Slider] [Explain]

[+ icon] [Copy icon] [Code icon] 25 1 - 5 of 5 [Refresh icon]

```
{
  "_id": ObjectId('6a81595b41ae4d61ab69273b'),
  "book_id": 102,
  "title": "Dune",
  "author": "Frank Herbert",
  "published_year": 2022,
  "category": "Science Fiction",
  "format": "Digital EPUB",
  "file_size": "3.1 MB",
  "member_id": 102,
  "reviews": Array (3)
}
```

STEP 5: Insert Document 3 in book_meta

Insert Document

To collection bookflow.book_metadata

VIEW

```
1 {
2   "book_id": 103,
3   "title": "Clean Code",
4   "author": "Robert C. Martin",
5   "published_year": 2019,
6   "category": "Programming",
7   "format": "Physical",
8   "dimensions": {
9     "height": "23 cm",
10    "width": "18 cm",
11    "weight": "700 g"
12  },
13   "member_id": 103,
14   "reviews": [
15     {
16       "member_id": 101,
17       "rating": 5,
18       "comments": "Excellent programming reference"
19     },
20     {
21       "member_id": 106,
22       "rating": 4,
23       "comments": "Very useful coding practices"
24     }
25   ]
26 }
27
```

Cancel

Insert

ORGANIZATION PENNETI's Org - 2026... PROJECT Project 0

Compare tiers

Data Explorer

- My Queries
- Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - customers
 - transactions
 - bookflow
 - book_metadata**
 - local

book_metadata bankdb

Cluster0 > bookflow > book_metadata View monitoring

Documents 5 Aggregations Schema Indexes 2

Search

25 1 - 5 of 5

```
{
  "_id": ObjectId('6a81595b41ae4d61ab69273c'),
  "book_id": 103,
  "title": "Clean Code",
  "author": "Robert C. Martin",
  "published_year": 2019,
  "category": "Programming",
  "format": "Physical",
  "dimensions": Object (3)
    member_id: 103
    reviews: Array (2)
```

STEP 6: Insert Document 4 in book_meta

Insert Document

To collection bookflow.book_metadata

VIEW

```
1 {
2   "book_id": 104,
3   "title": "Artificial Intelligence Basics",
4   "author": "John Smith",
5   "published_year": 2023,
6   "category": "Artificial Intelligence",
7   "format": "Digital PDF",
8   "file_size": "4.2 MB",
9   "member_id": 104,
10  "reviews": [
11    {
12      "member_id": 102,
13      "rating": 4,
14      "comments": "Good introduction to artificial intel
15    },
16    {
17      "member_id": 105,
18      "rating": 5,
19      "comments": "Excellent machine learning explanatio
20    },
21    {
22      "member_id": 106,
23      "rating": 4,
24      "comments": "Useful for beginners"
25    }
26  ]
27 }
```

Cancel

Insert

Navigation bar: Organization (PENNETH's Org - 2026...), Project (Project 0), and various utility icons.

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - bookflow
 - book_metadata**
 - local
 - sample_mflix

book_metadata +

Cluster0 > bookflow > book_metadata

View monitor

Documents 5 Aggregations Schema Indexes 1

Timeline controls: Clock icon, range slider, Explain button

Controls: +, Copy, Code icon, 25, 1 - 5 of 5, Refresh

```
_id: ObjectId('6a81595b41ae4d61ab69273d')
book_id: 104
title: "Artificial Intelligence Basics"
author: "John Smith"
published_year: 2023
category: "Artificial Intelligence"
format: "Digital PDF"
file_size: "4.2 MB"
member_id: 104
reviews: Array (3)
```

STEP 7: Insert Document 5 in book_meta

Insert Document

To collection `bookflow.book_metadata`

VIEW [Code icon] [Menu icon]

```
1 {
2   "book_id": 105,
3   "title": "Data Structures",
4   "author": "Mark Allen",
5   "published_year": 2024,
6   "category": "Computer Science",
7   "format": "Digital PDF",
8   "file_size": "5.0 MB",
9   "member_id": 105,
10  "reviews": [
11    {
12      "member_id": 101,
13      "rating": 5,
14      "comments": "Excellent data structures book"
15    },
16    {
17      "member_id": 107,
18      "rating": 3,
19      "comments": "Good but needs more examples"
20    }
21  ]
22 }
23
```

Cancel **Insert**

ORGANIZATION PENNETI's Org - 2026... PROJECT Project 0

Compare tiers ? [Menu] [User] [Bell] [Grid] [P]

Data Explorer

- My Queries
- Data Modeling

CLUSTERS (1)

Search clusters

- Cluster0
 - admin
 - bankdb
 - customers
 - transactions
 - bookflow
 - book_metadata**
 - local

book_metadata bankdb +

Cluster0 > bookflow > book_metadata View monitoring

Documents 5 Aggregations Schema Indexes 2 Vali

[Clock] [Slider] [Explain] [Reset]

[+][v] [Copy] [Code] 25 1 - 5 of 5 [Refresh] [Left] [Right]

```
_id: ObjectId('6a81595b41ae4d61ab69273e')
book_id: 105
title: "Data Structures"
author: "Mark Allen"
published_year: 2024
category: "Computer Science"
format: "Digital PDF"
file_size: "5.0 MB"
member_id: 105
reviews: Array (2)
```

STEP 8: "Comparsion" Operator \$gt

ORGANIZATION PENNETI's Org - 2026... PROJECT Project 0

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Search query: `{"reviews.rating":{"$gt":4}}`

Buttons: Explain, Reset, Find, Options

Results: 1 - 5 of 5

```
_id: ObjectId('6a81595b41ae4d61ab69273a')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)
```

STEP 9: Using \$IN

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Cluster0

admin

bankdb

bookflow

book_metadata

local

sample_mflix

book_metadata

+

Cluster0 > bookflow > book_metadata

Documents 5

Aggregations

Schema

Indexes 1

Validation

`{"reviews.rating":{"$in":[4,5]}}`

ADD DATA

BULK

EXPORT CODE

```
_id: ObjectId('6a81595b41ae4d61ab69273a')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)
```

```
_id: ObjectId('6a81595b41ae4d61ab69273b')
book_id: 102
title: "Dune"
author: "Frank Herbert"
published_year: 2022
```

STEP 10: Regex Query



ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
- bookflow
 - book_metadata
 - local
 - sample_mflix

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

{ "format": { "\$regex": "Digital", "\$options": "i" } }

ADD DATA BULK EXPORT CODE

```
_id: ObjectId('6a81595b41ae4d61ab69273a')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)
```

```
_id: ObjectId('6a81595b41ae4d61ab69273b')
book_id: 102
title: "Dune"
author: "Frank Herbert"
published_year: 2022
category: "Science Fiction"
format: "Digital EPUB"
file_size: "3.1 MB"
member_id: 102
reviews: Array (3)
```

AGGREGATION PIPELINE

STEP 11: \$Match

?

P

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

book_metadatadata

+

Cluster0 > bookflow > book_metadatadata

View monitoring

Visualize Your Data

Documents 5

Aggregations

Schema

Indexes 1

Validation

Smatch

✦

💡

Explain

Run

Options ▶

Untitled - modified

📁

+ CREATE NEW

</>

PREVIEW

{ }

</>

✦

⚙️

Stage 1 \$match

🔍

⋮

1 {

2 "published_year": {

3 "\$gt": 2020

4 }

5 }

Output preview after \$match

stage (Sample of 4 documents)

```
_id: ObjectId('6a81595')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fic
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
```

Step 12: \$unwind

ORGANIZATION PENNETI's Org - 2026... PROJECT Project 0

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 1 Validation

Smatch \$unwind

Untitled - modified CREATE NEW

PREVIEW

Stage 2 (\$unwind)

```
1 "$reviews"|
2
3
```

Output preview after \$unwind stage (Sample of 10 documents)

```
{
  "_id": ObjectId('6a81595...'),
  "book_id": 101,
  "title": "The Martian",
  "author": "Andy Weir",
  "published_year": 2021,
  "category": "Science Fic...",
  "format": "Digital PDF",
  "file_size": "2.5 MB",
  "member_id": 101
}
```

Alert System Status: All Good

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STEP 13: \$group

?

P

ORGANIZATION

PENNETI's Org - 2026...

>

PROJECT

Project 0

book_metadatadata

+

Cluster0 > bookflow > book_metadatadata

View monitoring

Visualize Your Data

Documents 5

Aggregations

Schema

Indexes 1

Validation

Smatch

Sunwind

\$group

Explain

Run

Options

Untitled - modified

+ CREATE NEW

</>

PREVIEW

</>

Stage 3

\$group

1 /**

2 * _id: The id of the group.

3 * fieldN: The first field name.

4 */

5 {

6 _id: "\$title",

7 average_rating: {

8 \$avg: "\$reviews.rating"

9 },

10 total_reviews: {

11 \$sum: 1

12 }

13 }

Output preview after \$group stage (Sample of 4 documents)

_id: "Artificial Intelligenc

average_rating: 4.333333333

total_reviews: 3

STEP 14: \$sort

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

book_metadatadata

Cluster0 > bookflow > book_metadatadata

View monitoring Visualize Your Data

Documents 5 Aggregations Schema Indexes 1 Validation

Smatch Sunwind \$group \$sort Explain Run Options

Untitled - modified CREATE NEW PREVIEW

Stage 4 \$sort

```
1 {
2   "average_rating": -1
3 }
```

Output preview after \$sort stage (Sample of 4 documents)

```
_id: "The Martian"
average_rating : 4.6666666666
total_reviews : 3
```

STEP 15: Expected Aggregation OUTPUT

ORGANIZATION
PENNETI's Org - 2026... ▾

PROJECT
Project 0 ▾

Compare tiers ? P

book_metadatadata +

Cluster0 > bookflow > book_metadatadata [View monitoring](#) [Visualize Your Data](#)

Documents 5 Aggregations Schema Indexes 1 Validation

Smatch Sunwind +2 Edit Explain Run Options ▸

ALL RESULTS `</>` Showing 1 - 4 count results < > ▾ ⋮ {}

<pre>_id: "Dune" average_rating: 4.666666666666667 total_reviews: 3</pre>
<pre>_id: "The Martian" average_rating: 4.666666666666667 total_reviews: 3</pre>
<pre>_id: "Artificial Intelligence Basics" average_rating: 4.333333333333333 total_reviews: 3</pre>
<pre>_id: "Data Structures" average_rating: 4 total_reviews: 2</pre>

STEP 16: Create Text Index

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

Compare tiers

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema **Indexes 2** Validation

Create Index Refresh [Manage your search indexes](#)

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR	36.9 kB	1 (since Sun Aug 16 2026)	UNIQUE	READY
> reviews.comments_text	TEXT	20.5 kB	0 (since Sun Aug 16 2026)		READY

STEP 17: Search using Text Index

ORGANIZATION PENNETI's Org - 2026... PROJECT Project 0

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
 - customers
 - transactions
- bookflow
 - book_metadata**
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

```
db.book_metadata.find(
  {
    "$text": {
      "$search": "excellent"
    }
  }
)
```

ADD DATA BULK EXPORT CODE

```
_id: ObjectId('6a81c2493dfb9f4537eba16c')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)
```

```
_id: ObjectId('6a81c2843dfb9f4537eba16e')
book_id: 102
```

STEP 18: USE Explain

ORGANIZATION PENNETI's Org - 2026... PROJECT Project 0

Data Explorer

My Queries
Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

- admin
- bankdb
 - customers
 - transactions
- bookflow
 - book_metadata**
 - local
 - sample_mflix

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

```
db.book_metadata.find(
  {
    "$text": {
      "$search": "excellent"
    }
  }
).explain("executionStats")
```

ADD DATA BULK EXPORT CODE

```
_id: ObjectId('6a81c2493dfb9f4537eba16c')
book_id: 101
title: "The Martian"
author: "Andy Weir"
published_year: 2021
category: "Science Fiction"
format: "Digital PDF"
file_size: "2.5 MB"
member_id: 101
reviews: Array (3)
```

```
_id: ObjectId('6a81c2843dfb9f4537eba16e')
book_id: 102
```

STEP 19: Explain Plain

Explain Plan

Explain provides key execution metrics that help diagnose slow queries and optimize index usage. [Learn more](#)

Visual Tree **Raw Output**

TEXT_MATCH
Returned **5** Execution Time **0 ms**

FETCH
Returned **5** Execution Time **0 ms**

IXSCAN
Returned **5** Execution Time **0 ms**
Index Name: **reviews.comments...**
Multi Key Index: **yes**

Query Performance Summary

- 5 documents returned
- 5 documents examined
- 0 ms execution time
- Is not sorted in memory
- 5 index keys examined

Query used the following index:

- _fts (text)**
- _ftsx ↑**

Close

STEP 20: Final Output

ORGANIZATION

PENNETI's Org - 2026...

PROJECT

Project 0

Data Explorer

My Queries

Data Modeling

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

bookflow

book_metadata

local

sample_mflix

book_metadata

+

Cluster0

bookflow

book_metadata

Documents

5

Aggregations

Schema

Indexes

2

Validation

Smatch

Sunwind

Sgroup

Ssort

Edit

ALL RESULTS

EXPORT CODE

_id: "Dune"

average_rating: 4.666666666666667

total_reviews: 3

_id: "The Martian"

average_rating: 4.666666666666667

total_reviews: 3

_id: "Artificial Intelligence Basics"

average_rating: 4.333333333333333

total_reviews: 3

_id: "Data Structures"

average_rating: 4

total_reviews: 2

ORGANIZATION

PENNETI's Org - 2026...

PROJECT

Project 0

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

bookflow

book_metadata

local

sample_mflix

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5 Aggregations Schema Indexes 2 Validation

{ "\$text": { "\$search": "excellent" } }

ADD DATA

BULK

EXPORT CODE

_id: ObjectId('6a81595b41ae4d61ab69273a')

book_id: 101

title: "The Martian"

author: "Andy Weir"

published_year: 2021

category: "Science Fiction"

format: "Digital PDF"

file_size: "2.5 MB"

member_id: 101

reviews: Array (3)

_id: ObjectId('6a81595b41ae4d61ab69273c')

book_id: 103

title: "Clean Code"

author: "Robert C. Martin"

published_year: 2019

category: "Programming"

format: "Physical"

dimensions: Object (3)

member_id: 103

reviews: Array (2)

ORGANIZATION

PENNETI's Org - 2026...

PROJECT

Project 0

CLUSTERS (1)

Search clusters

Cluster0

admin

bankdb

bookflow

book_metadata

local

sample_mflix

book_metadata

Cluster0

bookflow

book_metadata

Documents

5

Aggregations

Schema

Indexes

2

Validation

{"\$text":{"\$search":"excellent"}}

ADD DATA

BULK

EXPORT CODE

book_id : 105

title : "Data Structures"

author : "Mark Allen"

published_year : 2024

category : "Computer Science"

format : "Digital PDF"

file_size : "5.0 MB"

member_id : 105

reviews : Array (2)

_id : ObjectId('6a81595b41ae4d61ab69273d')

book_id : 104

title : "Artificial Intelligence Basics"

author : "John Smith"

published_year : 2023

category : "Artificial Intelligence"

format : "Digital PDF"

file_size : "4.2 MB"

member_id : 104

reviews : Array (1)

ORGANIZATION
PENNETI's Org - 2026...

PROJECT
Project 0

Compare tiers

book_metadata

Cluster0 > bookflow > book_metadata

Documents 5

Aggregations

Schema

Indexes 2

Validation

Create Index

Refresh

Manage your search indexes →

Name & Definition	Type	Size	Usage	Properties	Status
> _id_	REGULAR ⓘ	36.9 kB	1 (since Sun Aug 16 2026)	UNIQUE ⓘ	READY
> reviews.comments_text	TEXT ⓘ	20.5 kB	4 (since Sun Aug 16 2026)		READY